



Microprocessor & Computer Architecture (μ pCA)

UE24CS251

B

Session -

9



Microprocessor & Computer Architecture (μ pCA) UE24CS251 B

PSR & SWAP
Instructions

PSR Instructions

Move to Register from Status Register (MRS): Read from either CPSR or SPSR

Example:

```
MRS R0, CPSR
```

```
MRS R1, SPSR
```

Move to Status Register from Register (MSR):

_field

Example:

```
MSR CPSR_field, R0
```

```
MSR SPSR_field, R1
```

_c: Control Field (0:7)

_f: Flag Field(24:31)

_x: Extension (8:15)

_s: Status (16:23)

PSR Instructions

. text

MOVS R1, #- 10

MRS R0, CPSR

copy of the CPSR.

AND R0, R0, #0000

mode bits.

MSR CPSR_F, R0

the modified CPSR

. end

; Take a

; Clear the

; Write back

RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 00000000
R1	: ffffffff6
R2	: 00000000
R3	: 00000000
R4	: 00000000
R5	: 00000000
R6	: 00000000
R7	: 00000000
R8	: 00000000
R9	: 00000000
R10 (s1)	: 00000000
R11 (fp)	: 00000000
R12 (ip)	: 00000000
R13 (sp)	: 00005400
R14 (lr)	: 00000000
R15 (pc)	: 00001004

CPSR Register	
Negative (N)	: 1
Zero (Z)	: 0
Carry (C)	: 0
Overflow (V)	: 0
IRQ Disable	: 1
FIQ Disable	: 1
Thumb (T)	: 0
CPU Mode	: System

PSR Instructions

. text

MOVS R1, #- 10

MRS R0, CPSR

the CPSR

AND R0, R0, #0000

bits.

MSR CPSR_F, R0

modified CPSR

. end

; Take a copy of

; Clear the mode

; Write back the

RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 800000df
R1	: ffffffff6
R2	: 00000000
R3	: 00000000
R4	: 00000000
R5	: 00000000
R6	: 00000000
R7	: 00000000
R8	: 00000000
R9	: 00000000
R10 (s1)	: 00000000
R11 (fp)	: 00000000
R12 (ip)	: 00000000
R13 (sp)	: 00005400
R14 (lr)	: 00000000
R15 (pc)	: 00001008

CPSR Register	
Negative (N)	: 1
Zero (Z)	: 0
Carry (C)	: 0
Overflow (V)	: 0
IRQ Disable	: 1
FIQ Disable	: 1
Thumb (T)	: 0
CPU Mode	: System

0x800000df	

PSR Instructions

. text

MOVS R1, #- 10

MRS R0, CPSR

copy of the CPSR.

AND R0, R0, #0000

mode bits.

MSR CPSR_F, R0

the modified CPSR

. end

; Take a

; Clear the

; Write back



RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 00000000
R1	: ffffffff6
R2	: 00000000
R3	: 00000000
R4	: 00000000
R5	: 00000000
R6	: 00000000
R7	: 00000000
R8	: 00000000
R9	: 00000000
R10 (s1)	: 00000000
R11 (fp)	: 00000000
R12 (ip)	: 00000000
R13 (sp)	: 00005400
R14 (lr)	: 00000000
R15 (pc)	: 0000100c

CPSR Register	
Negative (N)	: 1
Zero (Z)	: 0
Carry (C)	: 0
Overflow (V)	: 0
IRQ Disable	: 1
FIQ Disable	: 1
Thumb (T)	: 0
CPU Mode	: System

0x800000df	

PSR Instructions

. text

MOVS R1, #- 10

MRS R0, CPSR ; Take a copy
of the CPSR.

AND R0, R0, #0000 ; Clear the mode
bits.

MSR CPSR_F, R0 ; Write back

the modified CPSR

. end

RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 00000000
R1	: ffffffff6
R2	: 00000000
R3	: 00000000
R4	: 00000000
R5	: 00000000
R6	: 00000000
R7	: 00000000
R8	: 00000000
R9	: 00000000
R10 (s1)	: 00000000
R11 (fp)	: 00000000
R12 (ip)	: 00000000
R13 (sp)	: 00005400
R14 (lr)	: 00000000
R15 (pc)	: 00001010

CPSR Register	
Negative (N)	: 0
Zero (Z)	: 0
Carry (C)	: 0
Overflow (V)	: 0
IRQ Disable	: 1
FIQ Disable	: 1
Thumb (T)	: 0
CPU Mode	: System

0x000000df	

SWAP : A & R1

RegistersView	
General Purpose Floating Point	
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 0
R1	: 5
R2	: 0
R3	: 0
R4	: 0
R5	: 0
R6	: 0
R7	: 0
R8	: 0
R9	: 0
R10 (s1)	: 0
R11 (fp)	: 0
R12 (ip)	: 0
R13 (sp)	: 21504
R14 (lr)	: 0
R15 (pc)	: 4100

```
.text
00001000:E3A01005    MOV R1,#5
00001004:E59F2008    LDR R2,=a
00001008:E5923000    LDR R3,[R2]
0000100C:E5821000    STR R1,[R2]
00001010:E1A01003    MOV R1,R3
.data
00001018:          a:.word 10
```

MemoryView1	
00001018	000A 0000 8181

SWAP : A & R1

RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 0
R1	: 5
R2	: 4120
R3	: 10
R4	: 0
R5	: 0
R6	: 0
R7	: 0
R8	: 0
R9	: 0
R10 (s1)	: 0
R11 (fp)	: 0
R12 (ip)	: 0
R13 (sp)	: 21504
R14 (lr)	: 0
R15 (pc)	: 4112

```
.text
00001000:E3A01005    MOV R1,#5
00001004:E59F2008    LDR R2,=a
00001008:E5923000    LDR R3,[R2]
0000100C:E5821000    STR R1,[R2]
00001010:E1A01003    MOV R1,R3
.data
00001018:                a:.word 10
```

MemoryView1			
00001018			
00001018	0005	0000	81
0000103C	8181	8181	81

SWAP : A & R1

RegistersView	
General	Step Over Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 0
R1	: 10
R2	: 4120
R3	: 10
R4	: 0
R5	: 0
R6	: 0
R7	: 0
R8	: 0
R9	: 0
R10 (s1)	: 0
R11 (fp)	: 0
R12 (ip)	: 0
R13 (sp)	: 21504
R14 (lr)	: 0
R15 (pc)	: 4116

CPSR Register	

```
.text
00001000:E3A01005    MOV R1,#5
00001004:E59F2008    LDR R2,=a
00001008:E5923000    LDR R3,[R2]
0000100C:E5821000    STR R1,[R2]
00001010:E1A01003    MOV R1,R3
.data
00001018:          a:.word 10
```

MemoryView1	
00001018	
00001018	0005 0000 8

Swap Instruction

SWP <Swap Destination>, <Original>,
[<address>]

Eg: SWP R0, R1, [R2]

.text

LDR R2, =a

MOV R1, #5

SWP R1, R1,

[R2]

.data

a: .word 10

RegistersView	
General Purpose	Floating Point
Hexadecimal	
Unsigned Decimal	
Signed Decimal	
R0	: 0
R1	: 10
R2	: 4112
R3	: 0
R4	: 0
R5	: 0
R6	: 0
R7	: 0
R8	: 0
R9	: 0
R10 (s1)	: 0
R11 (fp)	: 0
R12 (ip)	: 0
R13 (sp)	: 21504
R14 (lr)	: 0
R15 (pc)	: 70656

.text	
00001000:E59F2004	LDR R2,=a
00001004:E3A01005	MOV R1,#5
00001008:E1021091	SWP R1,R1,[R2]
.data	
00001010:	a:.word 10

After
Swapping

MemoryView1	
00001010	0005 0000 8181 8

1. Which instruction is used to read the contents of the Current Program Status Register (CPSR) or Saved Program Status Register (SPSR) into a general-purpose register?
- A. MSR B. SWP C. MRS D. LDR

2. When using the MSR instruction to write to a status register, specific fields can be targeted. According to the slides, which bit range does the Flag Field (`_f`) cover?
- A. Bits 0:7 B. Bits 8:15 C. Bits 16:23 D. bits24:31

1. Which instruction is used to read the contents of the Current Program Status Register (CPSR) or Saved Program Status Register (SPSR) into a general-purpose register?
- A. MSR B. SWP C. MRS D. LDR

Answer – C. MRS

2. When using the MSR instruction to write to a status register, specific fields can be targeted. According to the slides, which bit range does the Flag Field (_f) cover?
- A. Bits 0:7 B. Bits 8:15 C. Bits 16:23 D. Bits 24:31

Answer – D. Bits 24:32

3. Consider the following instruction : SWP R1, R1, [R2]. What operation does this instruction perform?

- A. It swaps the values inside registers R1 and R2.
- B. It loads the word from memory at address R2 into R1 and stores the original value of R1 into memory at address R2.
- C. It adds the value of R1 to the value at memory address R2.
- D. It compares the value in R1 with the value at memory address R2

4. To safely modify the bits in the CPSR, What is the correct logical order of operations shown?

- A. MSR (Write) → AND (Modify) → MRS (Read)
- B. MRS (Read) → AND (Modify) → MSR (Write)
- C. LDR (Load) → STR (Store)
- D. SWP (Swap) → MRS (Read)

3. Consider the following instruction : SWP R1, R1, [R2]. What operation does this instruction perform?

- A. It swaps the values inside registers R1 and R2.
- B. It loads the word from memory at address R2 into R1 and stores the original value of R1 into memory at address R2.
- C. It adds the value of R1 to the value at memory address R2.
- D. It compares the value in R1 with the value at memory address R2

Answer - B

4. To safely modify the bits in the CPSR, What is the correct logical order of operations shown?

- A. MSR (Write) → AND (Modify) → MRS (Read)
- B. MRS (Read) → AND (Modify) → MSR (Write)
- C. LDR (Load) → STR (Store)
- D. SWP (Swap) → MRS (Read)

Answer – B. MRS (Read) → AND (Modify) → MSR (Write)



THANK YOU



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